

EXERCISE 1: YOUR KOTLIN JETPACK COMPOSE MOBILE SHOP

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GitHub Repository: https://github.com/leo6300/COMP5450---Mobile-Programming_EXERCISE-1_Yingtian-Wu

Project Overview

Yingtian Wu's Sneaker Hub is a mobile shopping app built with Kotlin and Jetpack Compose. The app allows users to browse sneaker products, filter products by category and brand, search products, select shoe sizes, add items to a shopping bag, adjust quantities, remove items, and view the shopping bag total.

The project uses a simple in-memory list for product and cart data. No database is used, following the exercise requirement.

This project was developed using IntelliJ IDEA with Android and Jetpack Compose support, and it can also be opened as a full Android project in Android Studio.

Technologies Used

- Kotlin
- Jetpack Compose
- Material 3 UI components
- IntelliJ IDEA
- Android Emulator
- In-memory list for product and shopping bag data

Main Features

1. Welcome Screen: Displays the app name and a Get Started button.
2. Home Screen: Displays sneaker products in a two-column grid. Allows users to search sneakers by name, brand, or category. Allows users to filter sneakers by category. Allows users to filter sneakers by brand.

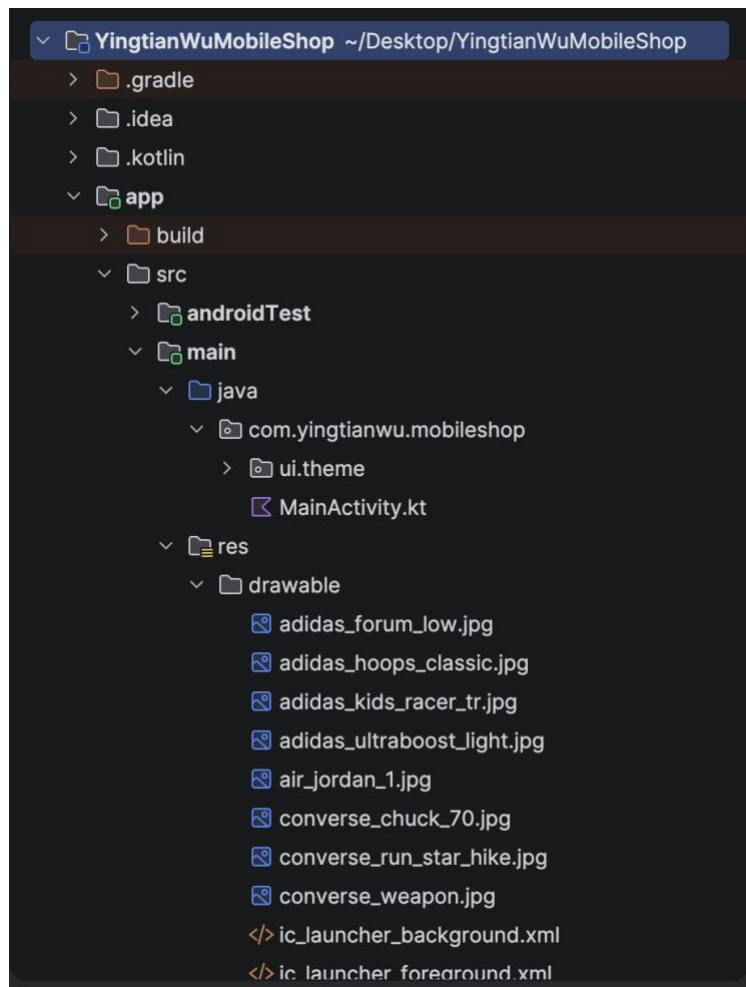
3. Product Detail Screen: Displays product image, name, brand, price, rating, and description. Allows users to select a shoe size. Allows users to add the selected product and size to the shopping bag.
4. Shopping Bag Screen: Displays selected products. Shows selected shoe size. Allows users to increase or decrease item quantity. Allows users to remove items. Displays the total shopping bag price.
5. Payment Screen: Displays the final total. Allows users to confirm cash payment. Clears the shopping bag after payment confirmation.

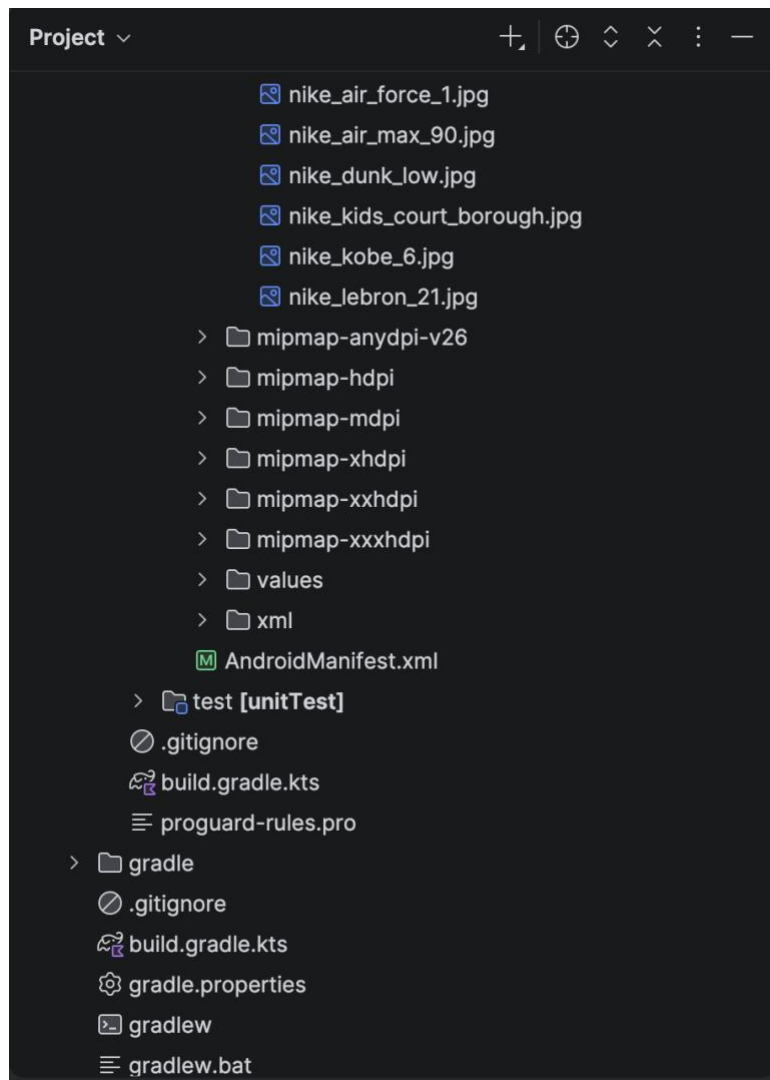
Project Structure

The project is organized as a full Android Kotlin Jetpack Compose project. The main Kotlin source file is located in `app/src/main/java/com/yingtianwu/mobileshop/MainActivity.kt`. The Compose theme files are stored in `ui.theme`. Product images are stored in `app/src/main/res/drawable`. `AndroidManifest.xml` and Gradle configuration files are included as part of the full Android project.

Main files and folders:

- `app/src/main/java/com/yingtianwu/mobileshop/MainActivity.kt`
- `app/src/main/java/com/yingtianwu/mobileshop/ui.theme/`
- `app/src/main/res/drawable/`
- `app/src/main/AndroidManifest.xml`
- `app/build.gradle.kts`
- `build.gradle.kts`
- `settings.gradle.kts`
- `gradle/`
- `gradlew`
- `gradlew.bat`





Important Kotlin Files / Components

- Product: Data class representing one sneaker product.
- CartItem: Data class representing one selected product in the shopping bag.
- Screen: Sealed class used for simple in-memory navigation.
- SneakerHubApp: Main composable that manages navigation, search, filters, and shopping bag state.
- WelcomeScreen: Starting screen of the app.
- HomeScreen: Displays product search, filters, product grid, and bag summary.
- ProductDetailScreen: Allows users to view product details and select a size.
- BagScreen: Displays cart items, quantity controls, remove button, and total price.

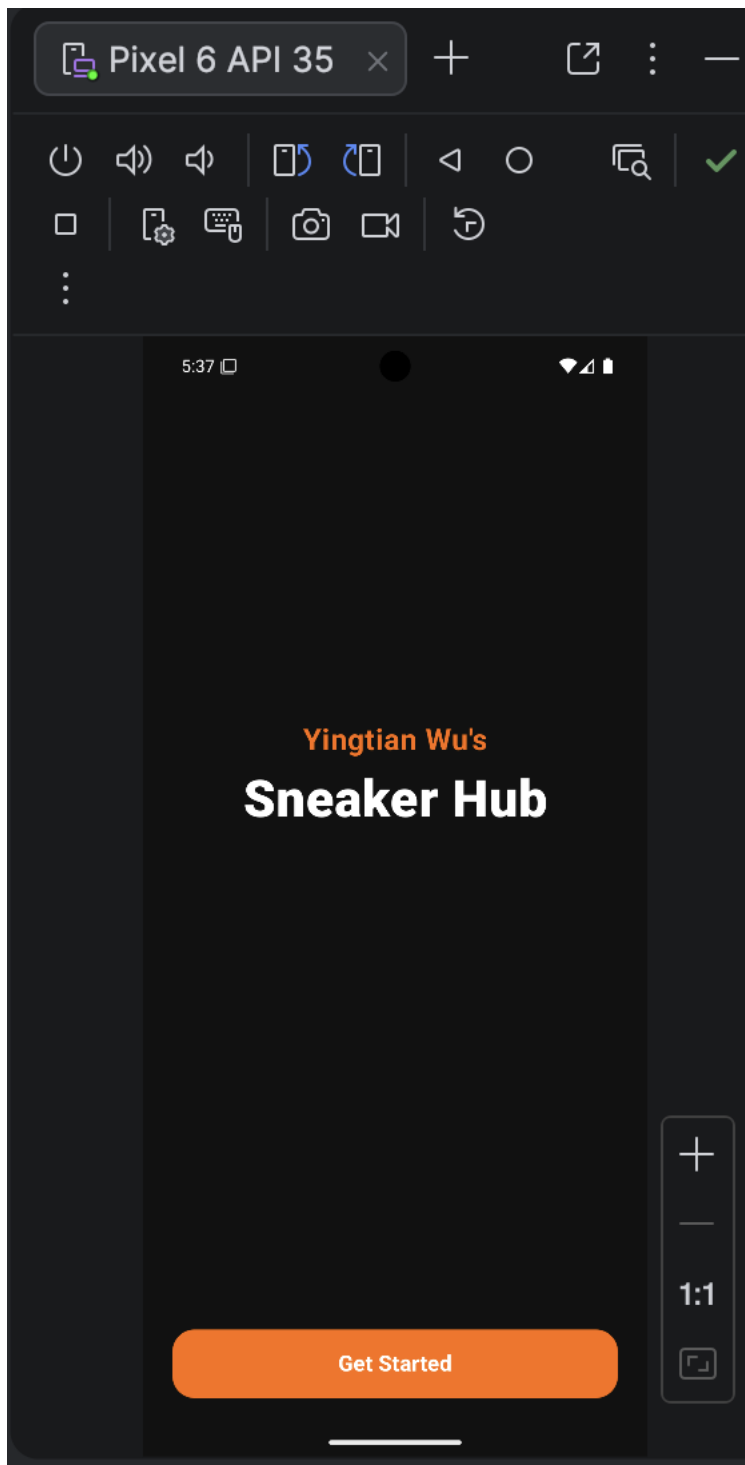
- PaymentScreen: Displays the final total and confirms payment.
- ProductCard: Reusable product card component.
- FilterRow: Reusable category and brand filter row.
- BagItemCard: Reusable shopping bag item component.

How to Configure and Run

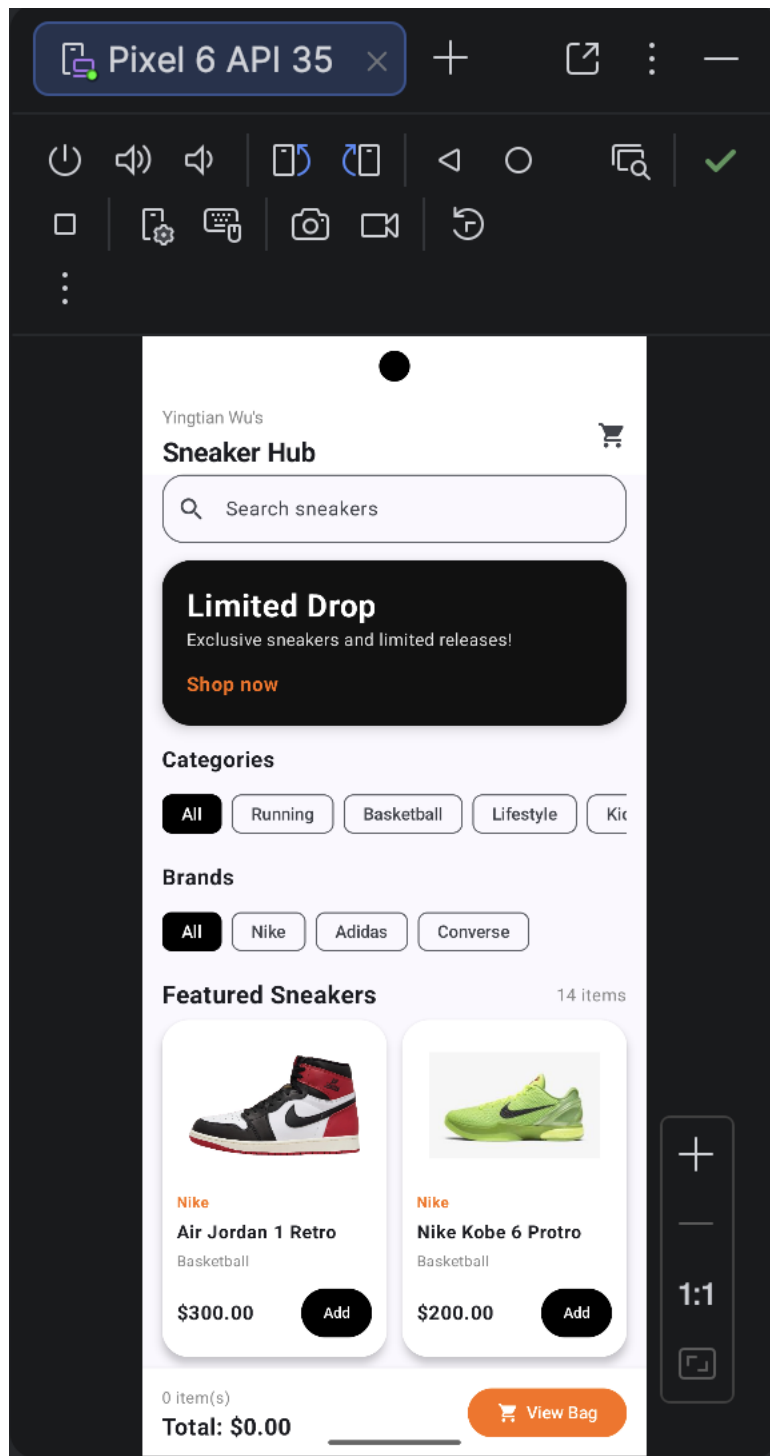
1. Download the project from GitHub or D2L.
2. Open IntelliJ IDEA or Android Studio.
3. Select File > Open.
4. Choose the project folder named "YingtianWuMobileShop".
5. Wait for Gradle Sync to finish.
6. Make sure the Android SDK is installed and configured.
7. Make sure an Android Emulator is available.
8. Select the emulator device.
9. Click the Run button.
10. The app will launch on the Android Emulator.

How to Use the App

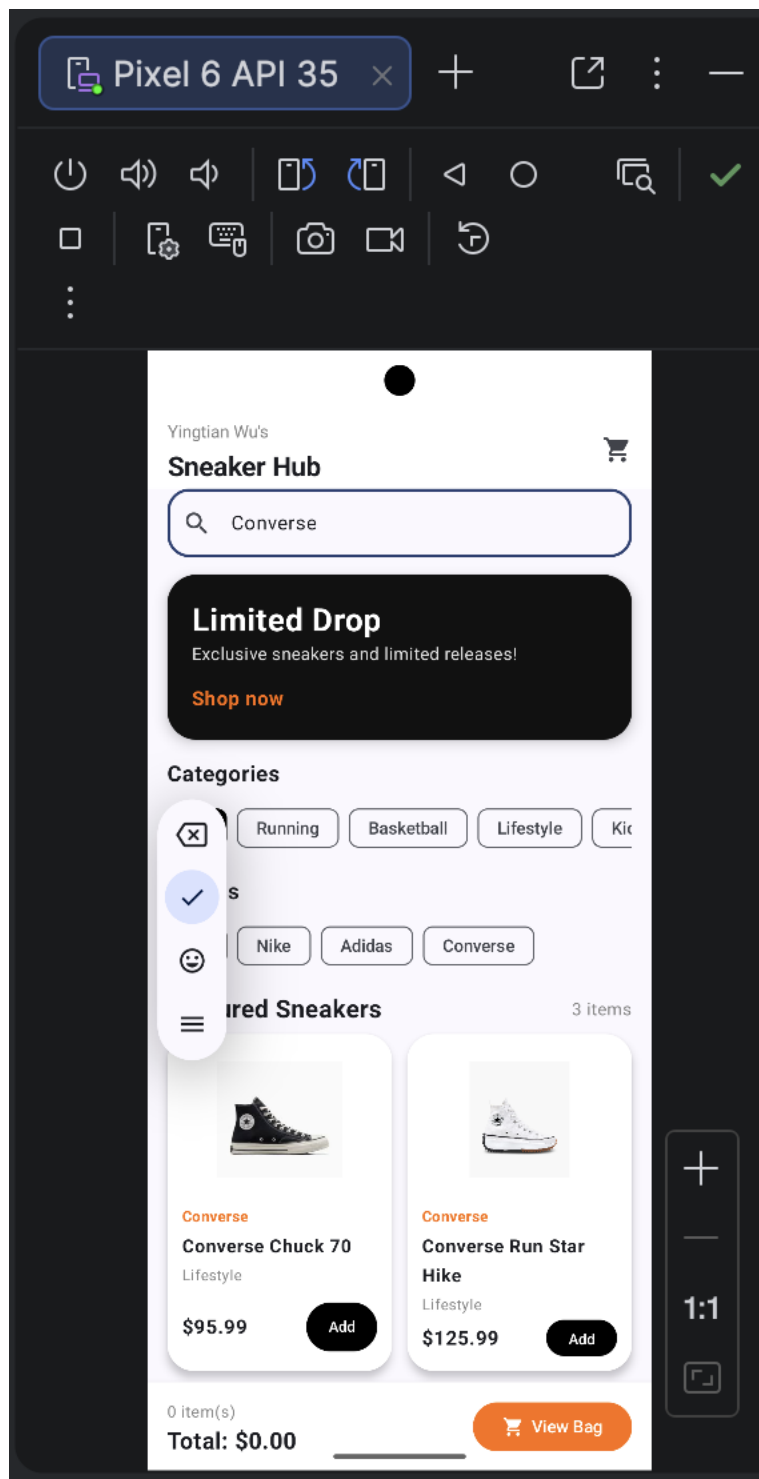
1. Run and Open the app.
2. Tap Get Started on the Welcome Screen.



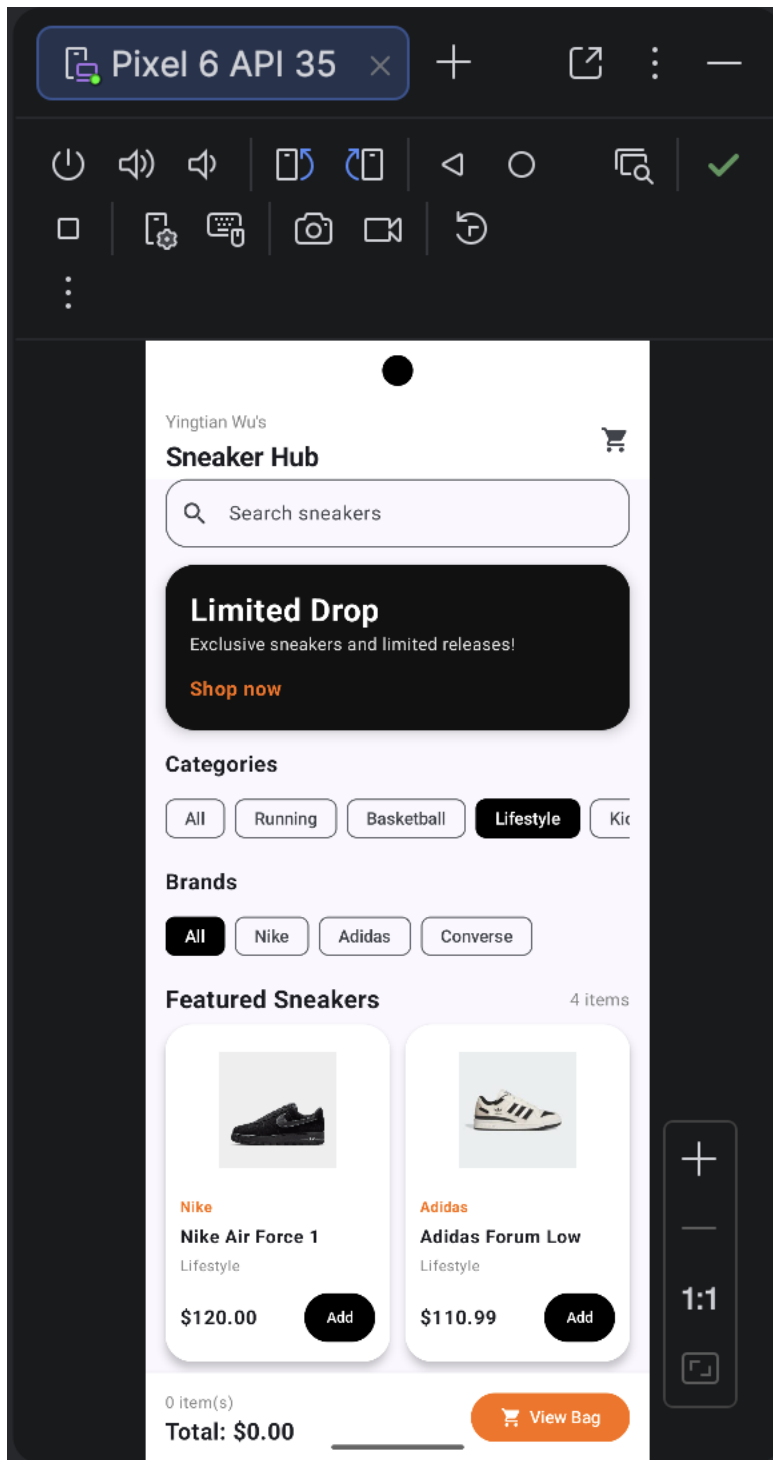
3. Browse sneakers on the Home Screen. Users can scroll up and down to view all available sneaker products.



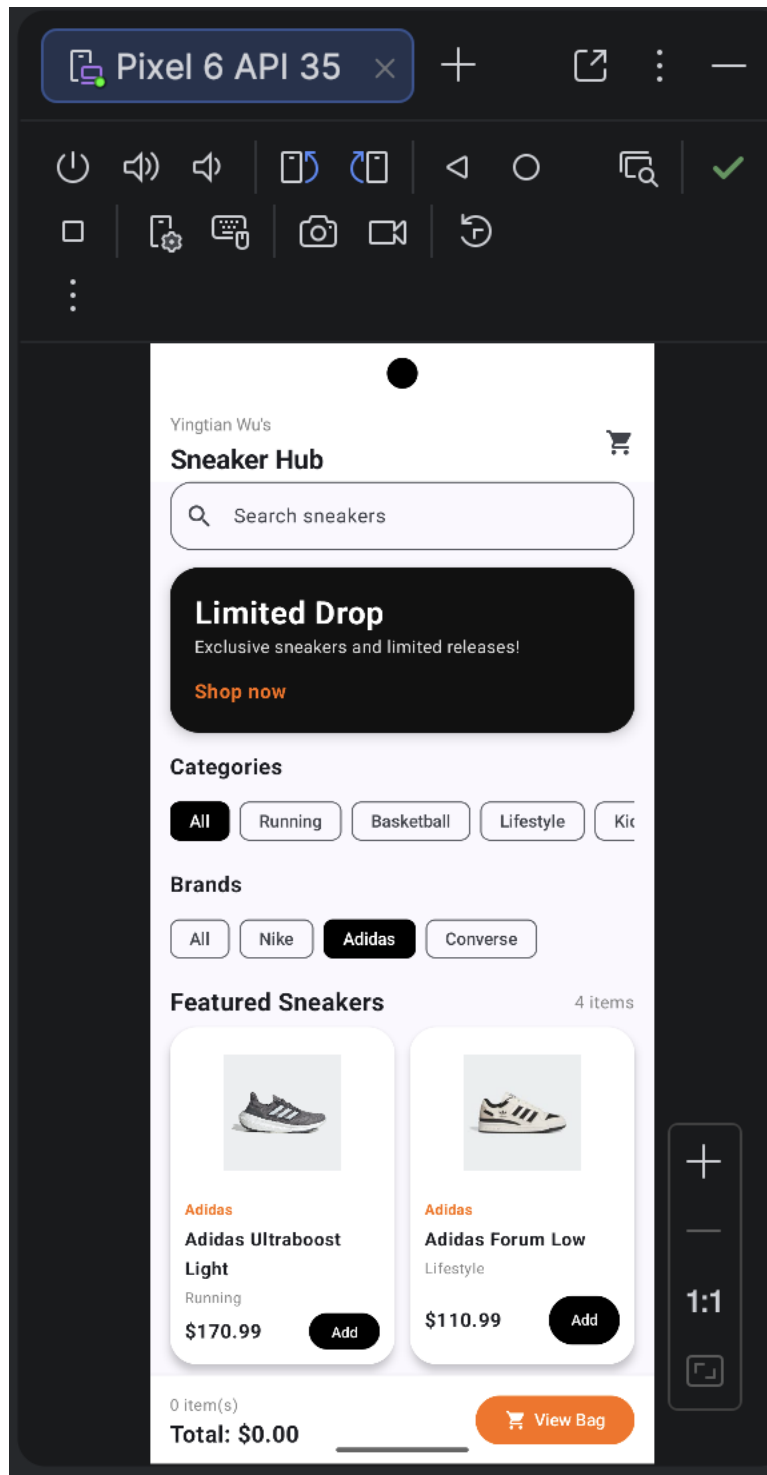
4. Use the search bar to search for sneakers by product name, brand, or category.



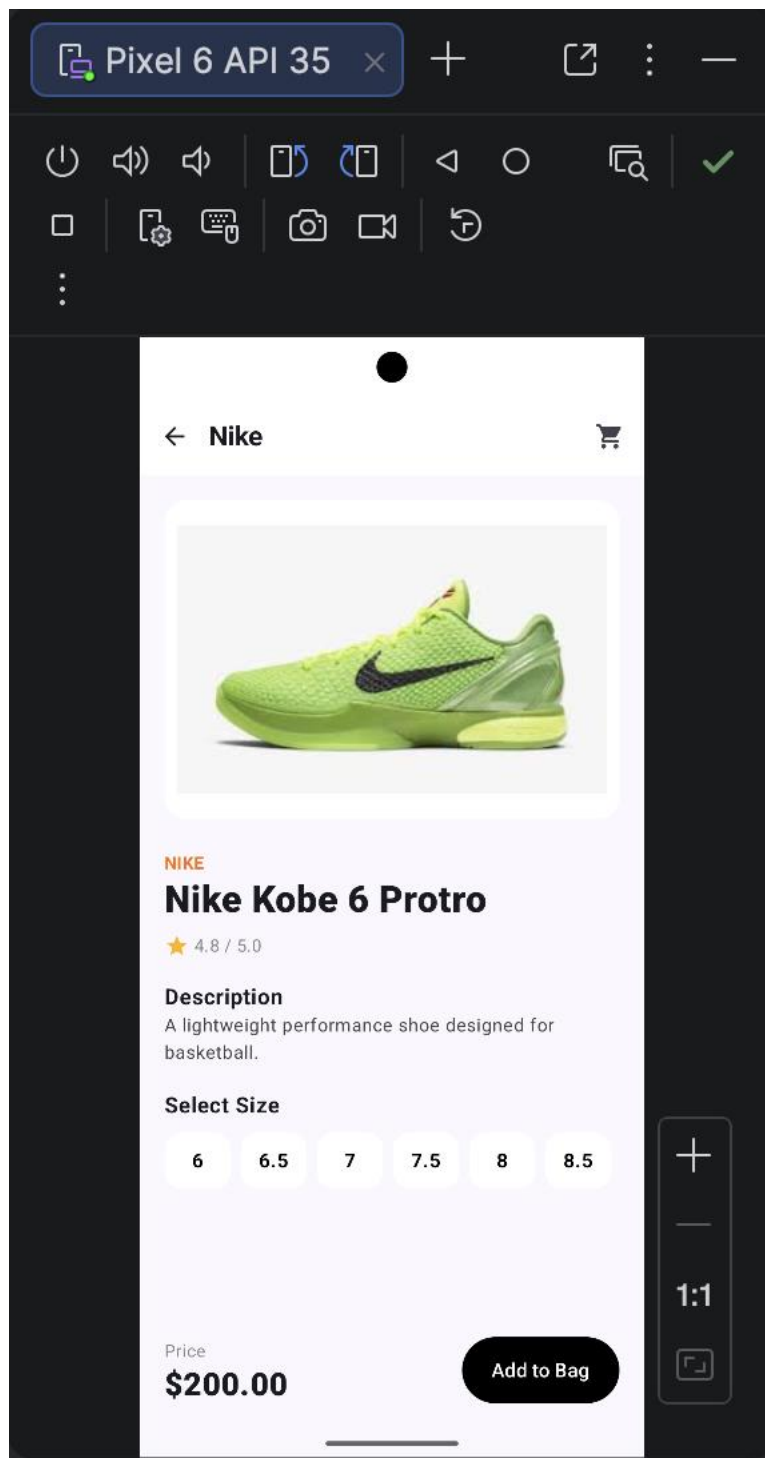
5. Select a category filter such as All, Running, Basketball, Lifestyle, or Kids to narrow the product list.



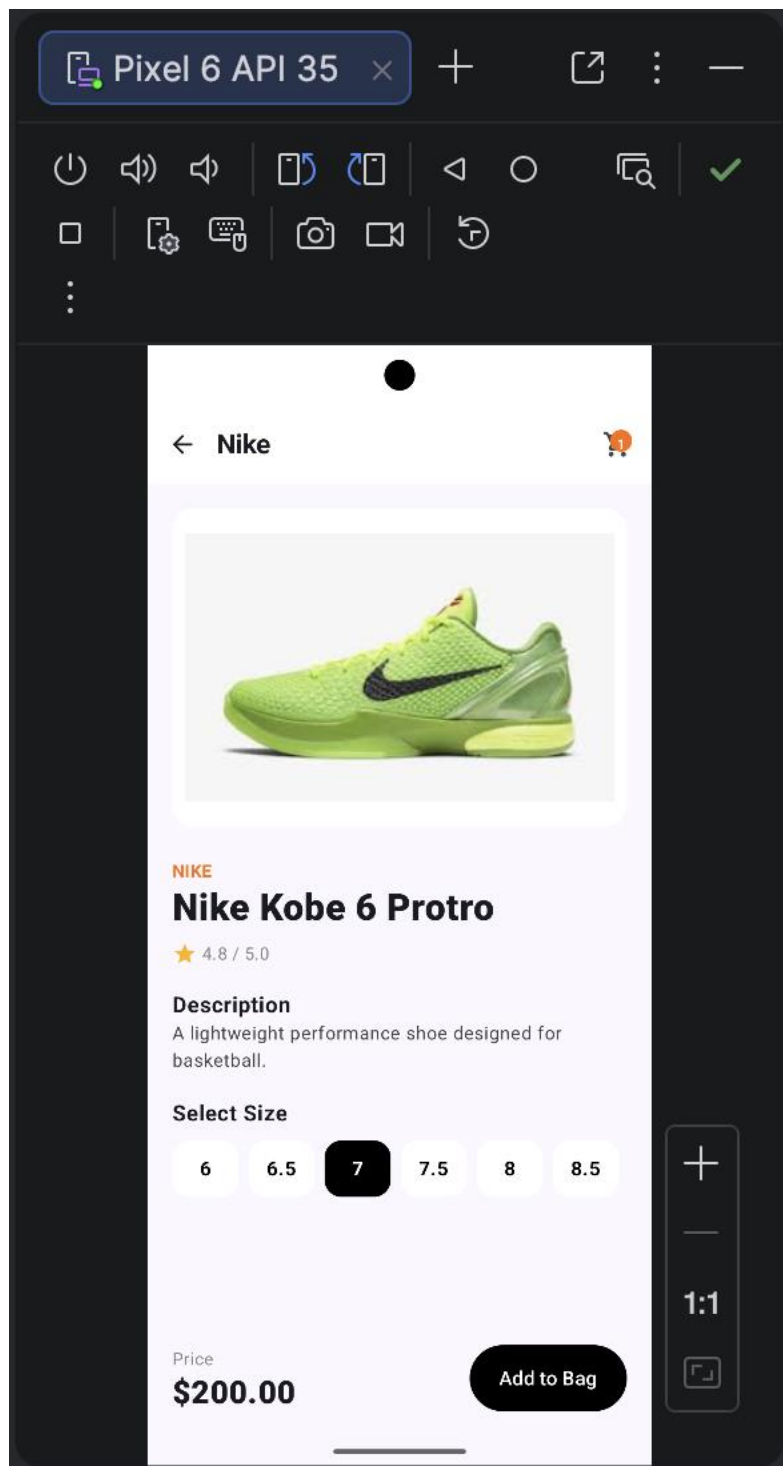
6. Select a brand filter such as All, Nike, Adidas, or Converse to view products from a specific brand.



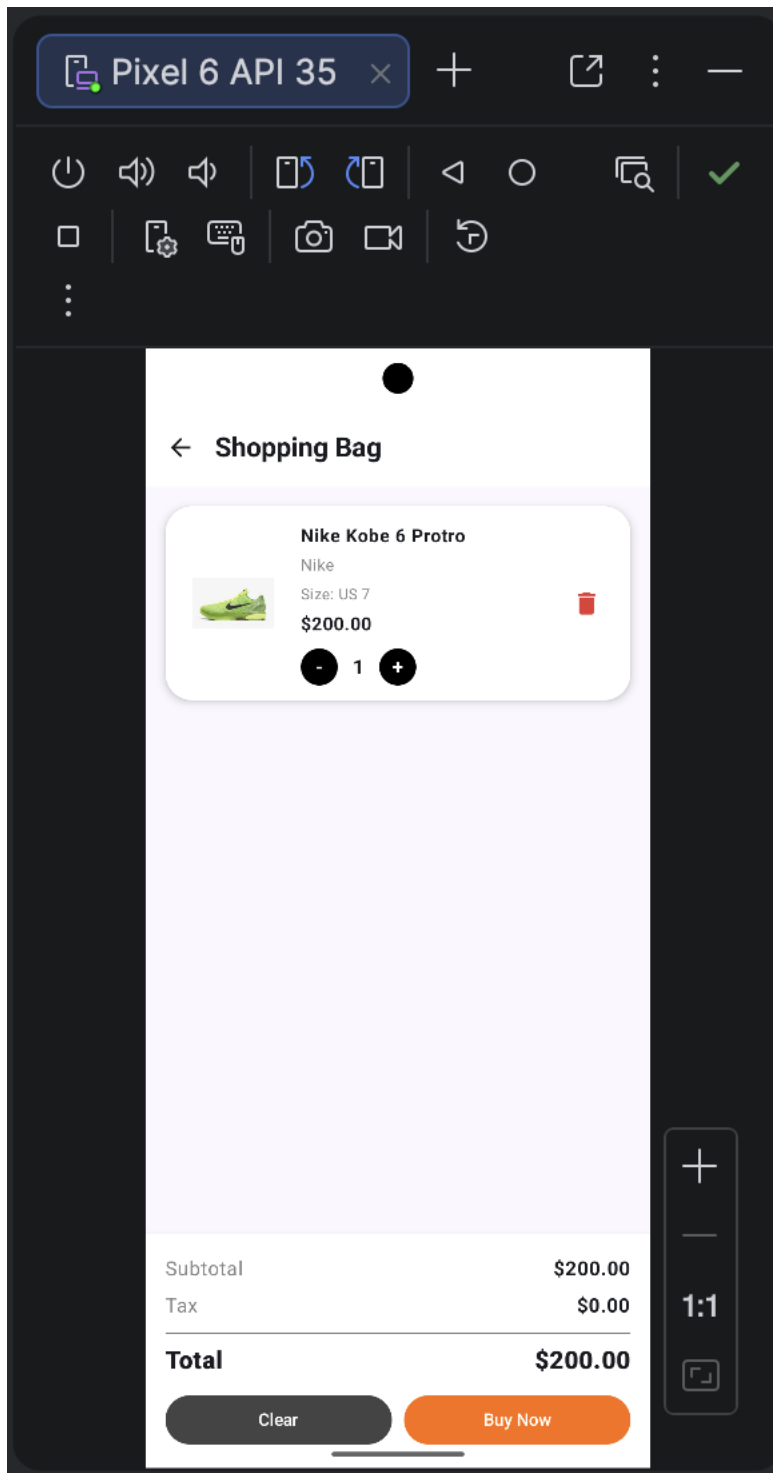
7. Tap a product card to open the Product Detail Screen.



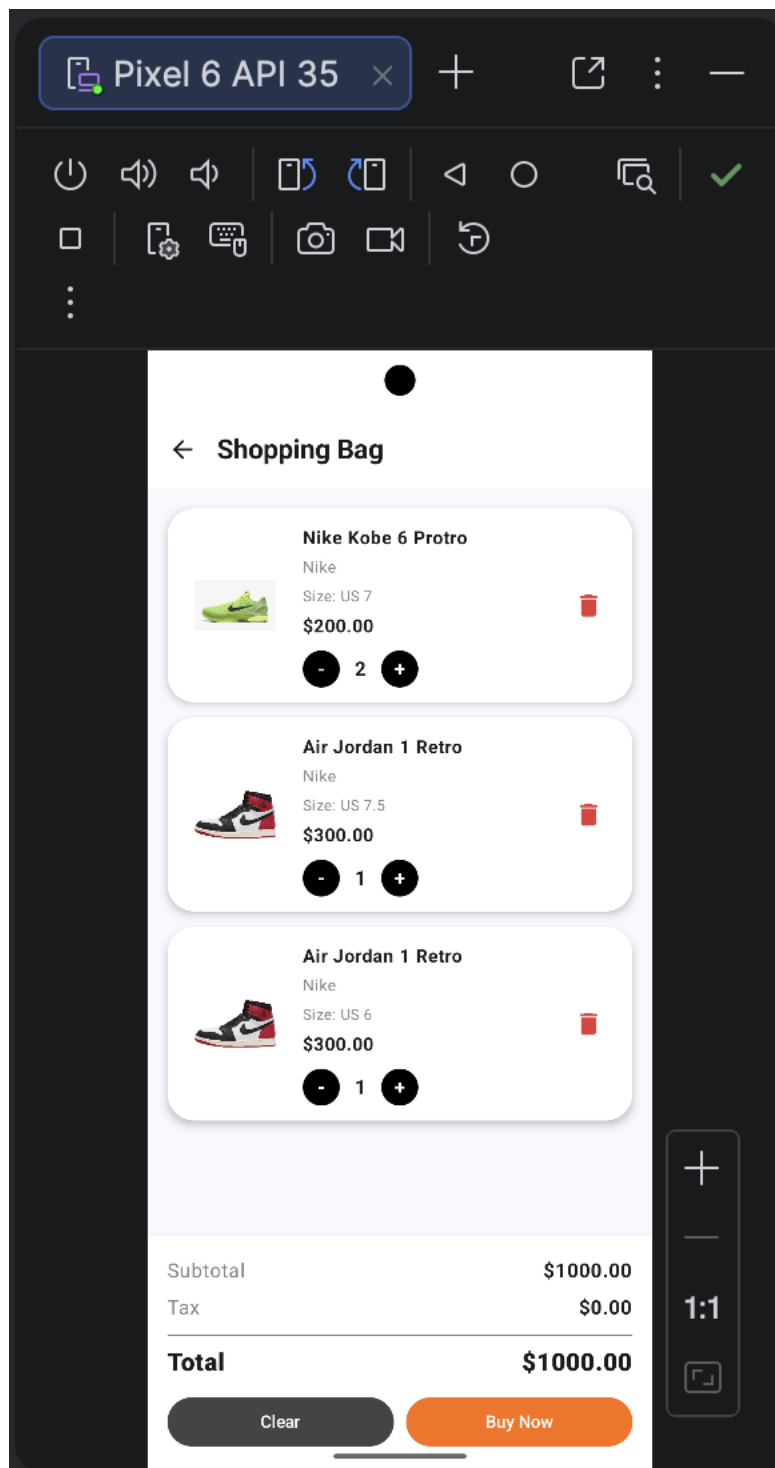
8. Select a shoe size and tap Add to Bag.



9. Tap the shopping bag icon in the top-right corner to open the Shopping Bag.



10. Increase, decrease, or remove items from the bag.



11. Tap Buy Now to open the Payment Screen.

